

1-2-1: Examining Network Standardization

At the end of this episode, I will be able to:

1. Identify and explain major network standardization organizations.

Learner Objective: *Identify and explain major network standardization organizations.*

Description: In this episode, the learner will examine various network standardization bodies and the roles these groups, organizations and consortiums that contribute to and promote network standards in the US, and internationally.

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- Network Standardization in the US and internationally
 - o International Organization for Standardization (ISO)
 - ♣ Develops and publishes international standards, including those for networking, like the OSI model.
 - ♣ Examples:
 - ♣ ISO/IEC 27000-series (Information Security Management Systems)
 - ♣ ISO/IEC 20000 (IT Service Management)
 - ♣ ISO/IEC 11801 (Cabling Standards for Networking)
 - ♣ ISO/IEC 7498-1 (Open Systems Interconnection model)
 - o The Internet Assigned Numbers Authority (IANA)
 - ♣ Coordinates key Internet functions such as IP address allocation, DNS root zone administration, and protocol parameter assignments, under the oversight of ICANN.
 - o Internet Corporation for Assigned Names and Numbers (ICANN)
 - ♣ Coordinates and maintain several databases related to the namespaces of the Internet, ensuring the network's stable and secure operation.
 - o Internet Engineering Task Force (IETF)
 - ♣ Develops and promotes voluntary Internet standards, particularly standards that comprise the Internet protocol suite (TCP/IP).
 - ♣ Examples:
 - ♣ RFC 791 (Internet Protocol, IP)
 - ♣ RFC 793 (Transmission Control Protocol, TCP)
 - ♣ RFC 2616 (Hypertext Transfer Protocol, HTTP)
 - ♣ RFC 1918 (Address Allocation for Private Internets)
 - o Institute of Electrical and Electronics Engineers (IEEE)
 - ♣ Develops international standards for various industries, including telecommunications, information technology, and power generation.
 - ♣ Examples:
 - ♣ IEEE 802.3 (Ethernet Standard for Wired LANs)
 - ♣ IEEE 802.11 (Wi-Fi Wireless LAN Standards)
 - ♣ IEEE 802.15.1 (Standard for Bluetooth Wireless Technology)
 - ♣ IEEE 802.1Q (VLAN Tagging Protocol)
 - o American National Standards Institute (ANSI)
 - ♣ Oversees the creation and dissemination of norms and standards in the United States.
 - ♣ Examples:
 - ♣ ANSI/ISA-95 - Enterprise-Control System Integration
 - ♣ ANSI C63.4 - Radio-Noise Emissions Measurement
 - ♣ ANSI/TIA-568 - Structured Cabling
 - ♣ ANSI X3.4-1968 - ASCII (American Standard Code for Information Interchange)

- Telecommunications Industry Association (TIA)
 - ♣ Develop standards for telecommunications products and services in the United States.
 - ♣ Examples:
 - ♣ TIA-568 - Commercial Building Telecommunications Cabling Standard
 - ♣ TIA-607 - Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
 - ♣ TIA-942 - Telecommunications Infrastructure Standard for Data Centers
 - o Federal Communications Commission (FCC)
 - ♣ Regulates interstate and international communications radio, television, wire, satellite, and cable in the United States.
 - o European Telecommunications Standards Institute (ETSI)
 - ♣ Produces internationally applicable standards for information and communications technologies (ICT), in Europe.
 - o International Telecommunication Union (ITU)
 - ♣ An agency of the United Nations concerned with information and communication technologies.
 - ♣ Examples:
 - ♣ ITU-T G.709 (Interfaces for the Optical Transport Network) *ITU-T G.9959 (Short-range wireless devices)
 - ♣ ITU-T H.264 (Advanced video coding for generic audiovisual services)
 - o World Wide Web Consortium (W3C)
 - ♣ An international group that developing open standards to ensure the long-term growth of the Web.
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- Additional Resources:
 - o If applicable